Fundamentals of Machine Learning

Understanding, Proofs and Experiments: Problems

Part I: understanding questions and proofs. Part II: real-world situations and experiments to read, diagnose and design.

How to use this set

Part I is about understanding and mathematics. It opens with eight short questions that each want a precise two-or-three-sentence answer, the kind an examiner uses to check whether a definition has been understood rather than memorised. Then four problems derive results the course uses without proving: why cross-entropy training produces probabilities, how much data model selection really needs, why LDA and Naive Bayes are secretly linear classifiers, and why no classifier is best at every prevalence.

Part II is about situations. Each problem gives you an experiment or a deployment the way it would arrive at work: a table of results, a protocol someone followed, a number someone reported. Your job is to read it, say what it does and does not show, compute what can be computed, and design the next experiment.

Each problem opens with a blue box holding the setup, then a yellow box listing the tools with a pointer [one-read §x] to the one-read document. Work on paper first; the same set with full solutions is *ML_Understanding_Proofs_Experiments_Solutions.pdf*. Difficulty: ⋆ direct, ⋆⋆ several steps, ⋆⋆⋆ derivation or design. All logarithms are natural.

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1 Part I: Understanding and proofs

1.1 I.0 Eight understanding questions

I.0 (⋆) **20 minutes**

Answer each in two or three sentences. Precision matters more than length.

(1) The posterior $p(y \mid x)$ is a property of the *problem*; the score $s_\theta(x)$ is a property of the *model*. Explain the distinction and say what connects them.

(2) Why can the Bayes error be strictly positive even with infinite training data and unlimited model capacity?

(3) The validation error of the *selected* model is a biased estimate of its population error, but its test error (used once) is not. Explain both halves.

(4) In what sense does the capacity of k-NN *decrease* as k grows, although the algorithm has no parameters?

(5) What does "generative models make their assumptions visible" mean concretely? Give one assumption and how it would be checked.

(6) Under prior shift, why does the posterior change while the class-conditional likelihoods do not?

(7) Why must preprocessing steps such as scaling and feature selection be fitted *inside* each cross-validation fold rather than once before cross-validation?

(8) A per-event classifier cannot detect a port scan, but a classifier on window features can. In what sense does the second one "see time"?

1.2 I.1 Which losses produce probabilities?

I.1 (⋆⋆) **15 minutes**

Fix an input x with true posterior $p = P(Y = 1 \mid x)$. A model outputs a number $r \in [0, 1]$ for this x. Consider three per-example losses and their *expected* value over Y:

$$\begin{aligned} \text{cross-entropy:} \quad & \ell_{\mathrm{CE}}(y, r) = -y \ln r - (1 - y) \ln(1 - r), \\ \text{squared:} \quad & \ell_{\mathrm{sq}}(y, r) = (y - r)^2, \\ \text{absolute:} \quad & \ell_{\mathrm{abs}}(y, r) = |y - r|. \end{aligned}$$

Tools you need

$\mathbb{E}_Y[\ell(Y, r)] = p\,\ell(1, r) + (1-p)\,\ell(0, r)$; entropy $H(p) = -p \ln p - (1-p)\ln(1-p)$ and KL divergence $\mathrm{KL}(p \,\|\, r) = p \ln \frac{p}{r} + (1 - p) \ln \frac{1-p}{1-r}$ [one-read §2.6].

(a) Show that the expected cross-entropy is minimised at $r = p$, and that $\mathbb{E}_Y[\ell_{\mathrm{CE}}(Y, r)] = H(p) + \mathrm{KL}(p \,\|\, r)$.

(b) Show that expected squared loss is also minimised at $r = p$.

(c) Show that expected absolute loss is minimised at $r \in \{0, 1\}$ (never at p unless $p \in \{0, \frac{1}{2}, 1\}$). What does this mean for a model trained with absolute loss?

(d) A colleague proposes training with the 0–1 loss on the thresholded prediction $\mathbf{1}(r \geq 0.5)$. Explain in one paragraph why this is both uninformative about probabilities and hard to optimise.

1.3 I.2 How much validation data does model selection need?

I.2 (⋆⋆⋆) 20 minutes

A fixed classifier f has population error $R(f)$. On n i.i.d. validation examples its empirical error is $\widehat{R}_n(f)$. Hoeffding's inequality for averages of n independent $\{0, 1\}$ variables states

$$P(|\widehat{R}_n(f) - R(f)| > \varepsilon) \leq 2e^{-2n\varepsilon^2}.$$

Tools you need

Union bound $P(A_1 \cup \cdots \cup A_M) \leq \sum_m P(A_m)$; generalisation gap and selection [one-read §3.1, 3.5]. Values: $\ln 40 = 3.689$, $\ln 4000 = 8.294$, $\ln 400{,}000 = 12.90$.

(a) For a single fixed classifier, how many validation examples guarantee $|\widehat{R}_n - R| \leq 0.02$ with probability at least 0.95?

(b) Now M candidate configurations $f_1, \ldots, f_M$ are evaluated on the *same* validation set and the best is selected. Show that with probability at least $1 - \delta$, *every* candidate satisfies $|\widehat{R}_n(f_m) - R(f_m)| \leq \varepsilon$ provided $n \geq \frac{\ln(2M/\delta)}{2\varepsilon^2}$. Conclude that the selected $\hat{f}$ satisfies $R(\hat{f}) \leq \min_m R(f_m) + 2\varepsilon$.

(c) Compute n for $M = 100$ and $M = 10{,}000$ with $\varepsilon = 0.02$, $\delta = 0.05$. Compare with (a) and comment on the growth.

(d) A team has $n = 2{,}000$ validation flows and compares $M = 100$ configurations. What ε does the bound guarantee at $\delta = 0.05$? Then they iterate: they look at the validation results, invent 100 new configurations inspired by what worked, and evaluate those too. Does the bound still apply with $M = 200$?

1.4 I.3 Three ways to get the same linear classifier

I.3 (⋆⋆⋆) 20 minutes

Logistic regression, LDA and Bernoulli Naive Bayes look like three different models. Show that all three produce a posterior of the form $P(y = 1 \mid x) = \sigma(w^\top x + b)$, and identify w and b for the two generative models.

Tools you need

Posterior log-odds $\ln \frac{P(1|x)}{P(0|x)} = \ln \frac{p(x|1)}{p(x|0)} + \ln \frac{\pi_1}{\pi_0}$; $\sigma(z) = 1/(1 + e^{-z})$ and $z = \ln \frac{\sigma(z)}{1-\sigma(z)}$ [one-read §4.2–4.5].

(a) Show that for any binary classifier, $P(y = 1 \mid x) = \sigma(z(x))$ where $z(x)$ is the posterior log-odds. Hence a model is "logistic-shaped" exactly when its log-odds is affine in x.

(b) LDA. With $p(x \mid c) = \mathcal{N}(\mu_c, \Sigma)$ in d dimensions, shared Σ, show that $z(x) = w^\top x + b$ with $w = \Sigma^{-1}(\mu_1 - \mu_0)$ and $b = -\frac{1}{2}(\mu_1^\top \Sigma^{-1}\mu_1 - \mu_0^\top \Sigma^{-1}\mu_0) + \ln \frac{\pi_1}{\pi_0}$.

(c) Bernoulli Naive Bayes. With binary features and $p(x_j = 1 \mid c) = \theta_{jc}$, show that $z(x) = \sum_j w_j x_j + b$ with

$$w_j = \ln \frac{\theta_{j1}(1-\theta_{j0})}{\theta_{j0}(1-\theta_{j1})}, \qquad b = \ln \frac{\pi_1}{\pi_0} + \sum_j \ln \frac{1-\theta_{j1}}{1-\theta_{j0}}.$$

(d) Since all three are linear, what differs between them? For each pair, say which you expect to win when (i) data are scarce and the generative assumption is roughly right, (ii) data are plentiful and the assumption is wrong, (iii) two features are near-duplicates.

1.5 I.4 No classifier is best at every prevalence

I.4 ($\star\star$) **15 minutes**

A classifier with recall ρ and false-positive rate ϕ is used on a population with attack prevalence π, costs C_{FN} per miss and C_{FP} per false alert. Two classifiers are available:

$$A: \ \rho = 0.90, \ \phi = 0.06; \qquad B: \ \rho = 0.70, \ \phi = 0.02; \qquad C_{FN} = 10, \ C_{FP} = 1.$$

Tools you need

Expected cost per flow as a function of the prevalence; recall and FPR do not depend on π [one-read §3.6].

(a) Show that the expected cost per flow is $C(\pi) = \pi(1-\rho)C_{FN} + (1-\pi)\phi\, C_{FP}$, a straight line in π. Write it for A, B, and for the two trivial rules "never alert" and "always alert".

(b) Find the prevalence at which A and B have equal cost, and say which is better on each side. Then find where A and B each stop beating "never alert".

(c) Sketch the four lines on $\pi \in [0, 0.1]$ and read off the lower envelope. What does the envelope represent?

(d) A and B are in fact the *same* scoring model at two thresholds. Explain why that makes the prevalence-dependent choice easy to implement, and what should be re-estimated when the prevalence changes.

2 Part II: Situations and experiments

2.1 II.1 Reading learning curves, with a labelling budget

II.1 (⋆⋆) 20 minutes

Three models were trained on nested subsets of a flow dataset. Each cell is *training error / validation error*, validation on a fixed held-out set of 5,000 flows.

n	250	1,000	4,000	16,000
logistic regression	0.18 / 0.21	0.19 / 0.20	0.19 / 0.195	0.19 / 0.192
5-NN	0.05 / 0.30	0.06 / 0.22	0.07 / 0.17	0.07 / 0.14
flexible model	0.00 / 0.33	0.01 / 0.24	0.02 / 0.16	0.03 / 0.12

Labelling costs money: the team can afford 16,000 more labelled flows (to reach 32,000) or none.

Tools you need

Generalisation gap, bias and variance as functions of sample size, underfitting and overfitting [one-read §3.1–3.3].

(a) For each model, describe the behaviour of the training error, the validation error and the gap as n grows. Diagnose each as bias-dominated or variance-dominated at $n = 16{,}000$.

(b) Extrapolate each model's validation error to $n = 32{,}000$ and justify the extrapolation. Which model should get the labelling budget?

(c) The training error of the flexible model *rises* with n (0.00 → 0.03). A colleague says this shows the model is getting worse. Respond.

(d) Rather than labelling 16,000 random flows, the team could label 16,000 flows chosen because the current flexible model is *uncertain* about them (posterior near 0.5). Give one reason this could help and one way it could hurt the validation measurement.

2.2 II.2 An ablation study finds a shortcut

II.2 (⋆⋆) 20 minutes

A flow classifier uses six features. The team runs an *ablation*: retrain with one feature removed at a time, same split, same hyperparameters. Validation error:

removed feature	none	bytes	duration	dst port	TCP flags	hour of day	source subnet
validation error	0.040	0.050	0.045	0.060	0.050	**0.210**	**0.090**

Later, the full model is evaluated on flows captured during a different week, in which the red team ran its attacks during office hours from several subnets: error 0.35. The model without *hour of day* scores 0.08 on that same week.

Tools you need

Shortcuts versus leakage, data-collection artefacts, experiments that distinguish them [one-read §1.5, 3.6].

(a) Interpret the ablation table on its own. Which features does the model rely on most? Is anything suspicious from the table alone?

(b) Explain what the second week shows, and why the ablation on the original split could not have revealed it by itself.

(c) Is the hour feature target leakage? Argue carefully.

(d) Design two experiments, using only the *original* data, that would have raised the alarm before the second week.

(e) The team decides to keep the hour feature "because attacks really are more common at night in production". Under what conditions is that defensible, and what must they measure?

2.3 II.3 Labels that come only from alerts

II.3 ($\star\star\star$) **25 minutes**

A deployed detector processes 100,000 flows per day and alerts on about 2,000. Analysts investigate every alert and record a verdict: on a typical day 300 alerts are confirmed attacks. The verdicts are stored and used as labels to retrain the model every month. Nobody looks at the 98,000 un-alerted flows.

Tools you need

Precision and recall from counts; what a label process observes and what it misses [one-read §1.3, 3.6]; Poisson rule of thumb: a count k has standard deviation about $\sqrt{k}$.

(a) Which metrics can be computed from the analysts' verdicts, and which cannot? Compute what can be.

(b) Describe what goes wrong when the monthly retraining uses only these verdicts as training data.

(c) A random audit labels 1,000 of the un-alerted flows and finds 4 attacks. Estimate the daily number of missed attacks, the recall, and the prevalence. Give a rough uncertainty on each.

(d) How large should the audit be to estimate the number of missed attacks to within about $\pm 25\%$? Propose a sampling design that reaches that precision more cheaply than uniform random sampling.

(e) Design a labelling and retraining protocol that fixes (b). State what is labelled, how the training set is weighted, and how the next model is evaluated.

2.4 II.4 When the cost depends on the input: card fraud

II.4 (⋆⋆) **15 minutes**

A card-fraud model outputs a calibrated posterior p that a transaction is fraudulent. Blocking a legitimate transaction costs a fixed $C_{FP} = 5$ (customer friction). Letting a fraudulent transaction through costs its amount a, which is a *feature* of the transaction. Four transactions:

	T_1	T_2	T_3	T_4
amount a	20	500	5	80
posterior p	0.25	0.02	0.40	0.05

Tools you need

Bayes decision with input-dependent costs: compare $R(\text{block} \mid x) = C_{FP}(1-p)$ with $R(\text{allow} \mid x) = a\,p$ [one-read §2.2].

(a) Derive the threshold as a function of the amount, $t(a)$, and decide the four transactions.

(b) A colleague wants to add the amount as a feature to the model instead, keep a single global threshold of 0.5, and let the model "learn" that large amounts matter. Explain why this does not achieve the same thing.

(c) Fraud losses are reimbursed by the card network above $a > 200$, so the bank's cost of a missed large fraud is capped at 200. Redo T_2. Then explain why the model did not need retraining.

(d) Customer friction is not really constant: for a customer who has been blocked twice this month, a third block costs 25 (they may leave). Where in the system does this belong, and what data would you need to set it?

2.5 II.5 Aggregate metrics hide where the model fails

II.5 (⋆⋆) **15 minutes**

A detector is evaluated on one day of traffic from two network segments:

segment	flows	attacks	recall	FPR
office workstations	90,000	100	0.90	0.01
server farm	10,000	400	0.50	0.01

The report to management says: "overall false-positive rate 1%, overall accuracy 98.8%".

Tools you need

Counts from rates, aggregation across groups, per-group thresholds [one-read §3.6].

(a) Compute TP, FP and precision for each segment, then the overall recall, precision, FPR and accuracy. Check the report's numbers.

(b) Explain, in terms of where the attacks and the benign traffic sit, why the overall recall is 0.58 although one segment has recall 0.90, and why the office precision is so low although its recall is high.

(c) Analysts complain that "nine out of ten office alerts are noise" while security complains that "half the server attacks get through". Propose a change that addresses both without retraining, and say what it costs.

(d) Generalise: what should every evaluation report include besides the overall metrics, and why does accuracy appear in the report at all?

2.6 II.6 Monitoring a model after deployment

II.6 (⋆⋆⋆) **20 minutes**

A calibrated detector was deployed in January with a cost-based threshold of 0.10. A random audit of 2,000 flows per month provides labels. Monthly figures:

	Jan	Feb	Mar	Apr	May	Jun
mean predicted posterior on audit	0.012	0.012	0.013	0.013	0.014	0.014
attack rate in audit	0.012	0.011	0.014	0.019	0.025	0.031
recall on audit	0.82	0.80	0.74	0.63	0.55	0.47
FPR on audit	0.015	0.015	0.014	0.014	0.014	0.013

Tools you need

Calibration on average, prior shift versus concept drift, time-ordered validation [one-read §3.6, 4.7].

(a) Describe what is drifting and what is not. Is this prior shift, a change in $p(x \mid y)$, or both? Use the rows to argue.

(b) In June, what fraction of the audited attacks are missed, and about how many attacks per month are missed in production if there are 3 million flows per month? What is the model's calibration error in June?

(c) Two fixes are proposed: (i) apply the prior-shift correction with $\pi_d = 0.031$ and keep the model; (ii) retrain on January–June data. Say what each fixes and what it cannot fix.

(d) Design the monitoring protocol you would have wanted from January: what to compute monthly from the audit, what thresholds trigger action, and how to validate a retrained model before it replaces the old one.